

**TECHNICAL DATA SHEET**

# Hospital Management System (HMS)

Patient Engagement Platform | Electronic Patient Flow Management

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Our Hospital Management System (HMS) connects clinical, administrative and patient-facing operations on a single platform. This data sheet covers two core modules of the HMS suite — the Patient Engagement Solution (web portal and mobile app) and the Electronic Patient Flow Management (ePFM) system — along with the AI-powered capabilities and hospital-wide features that come with the platform.

## 1. Patient Engagement Solutions

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The Patient Engagement module gives patients secure, self-service access to their care journey — from booking an appointment to reviewing lab results and settling bills — without needing to visit the hospital in person. It is delivered as a web portal and a companion mobile app, both connected in real time to the hospital's core HMS for doctor schedules, appointments and lab reports.

### 1.1 Core Capabilities

- Book, reschedule and cancel appointments online
- View doctor schedules and availability
- View health information such as lab results and clinical notes
- View bills and make payments online
- Check prescriptions
- Schedule and track follow-up visits
- Receive health-related notifications
- Update personal medical history
- Send secure messages to care providers

### 1.2 Web Portal

- Secure, individual patient portal accounts
- Pre-registration for patients without online payment, with demographic details available to hospital staff on arrival
- Full registration with online payment and automatic hospital ID (UHID) generation
- Linking to existing medical records for returning patients
- Appointment booking, rescheduling and cancellation
- Access to past visit history, active medications and lab results

### 1.3 Mobile App (iOS & Android)

- All web portal features, optimised for mobile use
- Single login to manage the full family's health records
- OTP- or password-based login
- Runs on the same application server as the web portal, keeping data consistent across channels

### 1.4 AI-Powered Capabilities

The Patient Engagement module supports the following AI-driven features:

| AI Capability                           | What it does                                                                                                                                                    | Status    |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| AI Prescription & Lab Report Summarizer | Explains lab results and prescriptions in plain language, flagging values outside the normal range and summarising medication purpose, dosage and side effects. | Available |
| AI Medical History Voice Update         | Patients speak their health history — symptoms, past surgeries, allergies — and the AI extracts structured data to auto-fill the history form.                  | Available |
| AI Symptom Checker (Triage Assistant)   | Guides patients through a structured symptom conversation and recommends the right OPD department, with a pre-filled triage note sent ahead to the department.  | Available |
| AI Patient Support Chatbot              | 24/7 chatbot on web and mobile answering questions on appointments, lab reports, bills and doctor schedules, with voice support.                                | Roadmap   |
| AI WhatsApp Appointment Booking         | Book, reschedule or cancel appointments through WhatsApp using natural language, with automatic reminders.                                                      | Roadmap   |
| AI Pre-Registration Assistant           | A conversational flow that walks patients through pre-registration and securely sends demographic data to hospital staff.                                       | Roadmap   |

"Available" features are supported out of the box. "Roadmap" features are planned for future release.

## 2. Electronic Patient Flow Management (ePFM)

ePFM manages patient movement through OPD counters using electronic token dispensers, LED master displays, counter displays, self-service kiosks and a central server — all integrated with the HMS appointment module. It handles both walk-in patients and those with a prior appointment, and can organise queues by patient category (General, Ladies, Hospital Staff, Senior Citizen/Handicap) so waiting is fair and orderly.

### 2.1 What's Included

- Electronic Token Dispenser Units with touchscreen
- Master Display screens (LED TVs) showing live queue status
- Counter Displays at each registration/consultation point
- Self-service Kiosks with automatic token printing
- Central Server running the Manager Console software
- Full integration with the HMS appointment module, covering both walk-in and pre-booked patients

### 2.2 AI-Powered Capabilities

The ePFM module supports the following AI-driven features, planned as part of the product roadmap:

| AI Capability                           | What it does                                                                                                                                                                               | Status  |
|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| AI Voice & WhatsApp Appointment Booking | Patients book, confirm or reschedule OPD appointments using natural spoken language or WhatsApp, with real-time slot checking.                                                             | Roadmap |
| Conversational AI Kiosk Assistant       | Guides patients through check-in on the self-service kiosk by touch or voice, confirms appointments or issues a walk-in token, and prints the slip. Supports Hindi and regional languages. | Roadmap |
| AI Predictive Wait-Time Display         | Uses historical and real-time queue data to predict wait times per                                                                                                                         | Roadmap |

| AI Capability                 | What it does                                                                                                                                  | Status  |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|---------|
|                               | counter/doctor, shown on the LED master displays and sent to patients via WhatsApp.                                                           |         |
| AI Voice Patient Registration | Registration staff or patients speak demographic details aloud; the system auto-fills the registration form and checks for duplicate records. | Roadmap |

## 2.3 Indicative Hardware Specifications

| Component          | Specification                                                                                               |
|--------------------|-------------------------------------------------------------------------------------------------------------|
| Self-service Kiosk | 23" touchscreen with thermal printer (80mm x 58mm) and auto-cutter                                          |
| Application Server | Quad-core Xeon processor, 32 GB DDR4 RAM, 240 GB SSD + 1 TB HDD, integrated Gigabit LAN (wireless optional) |
| Master Display     | 50" smart TV / LED display                                                                                  |
| Secondary Display  | 40" smart TV / LED display                                                                                  |
| Counter Display    | 17" LED/LCD touchscreen, SAW touch technology, 3mm anti-glare tempered glass                                |

Quantities and screen placement are configured to match each hospital's OPD layout and scale.

## 3. Hospital-Wide Platform Features

Beyond patient engagement and flow management, the HMS platform includes the following capabilities:

### 1. Real-Time Hospital Control Towers

Live management dashboards for leakage-proof, real-time monitoring of operations, billing, pharmacy, inventory and clinical activities.

### 2. Prescription-Done-but-Not-Billed Tracking

Automatically identifies OP prescriptions generated by doctors but not billed in pharmacy, helping reduce revenue leakage.

### 3. Predictive Stock Management

AI-based stock prediction using patient flow and consumption patterns to calculate how much stock is required and for how long.

### 4. Smart Near-Expiry Control Tower

Automatically tracks near-expiry medicines and alerts departments before 90 days for timely return, replacement or credit-note processing.

### 5. Automated Department-Wise Revenue Analytics

Live revenue tracking across OP, IP, Lab, Radiology, Pharmacy, OT, ICU and specialty units.

### 6. Paperless Hospital Ecosystem

Dedicated mobile apps for patients, doctors, nurses and staff to minimise paper use and improve efficiency.

### 7. Automated NABH Quality Indicators

Quality indicators are auto-calculated directly from transactions and clinical data, without manual entry or spreadsheets.

### 8. Predictive Patient Flow Intelligence

Forecasts OP/IP patient loads from historical trends to help plan staffing and resources.

### 9. Clinical Leakage Monitoring

Tracks services ordered but not executed or billed, so diagnostic, pharmacy and procedure revenue is not missed.

**10. Doctor Productivity Dashboards**

Real-time view of consultation counts, procedures, surgery revenue, turnaround time and specialty performance.

**11. Smart OT Utilisation Analytics**

Tracks OT occupancy, surgery delays, idle time, surgeon utilisation and cancellation analytics.

**12. Automated Insurance and TPA Tracking**

Monitors pending approvals, query ageing, claim status and settlement timelines to improve cash flow.

**13. Centralised Multi-Branch Management**

Manage multiple hospitals, clinics and pharmacies from a single platform with consolidated reporting.

**14. Live Bed Occupancy & Discharge Prediction**

Real-time bed management with expected-discharge predictions to optimise admissions and occupancy planning.

**15. Integrated Biomedical Maintenance Alerts**

Tracks AMC, calibration schedules, breakdowns and preventive maintenance of biomedical equipment.

**16. Smart Nursing Monitoring System**

Tracks nursing activities, medication administration, vitals completion and pending assessments in real time.

**17. Automated Procurement Intelligence**

Suggests optimal purchase quantities based on consumption trends, vendor lead time and stock movement.

**18. Management Exception Alerts**

Instant alerts for abnormal discounts, excessive consultations, pending bills, stock variance or unusual activity.

**19. Digital Patient Engagement Platform**

Patients access appointments, prescriptions, lab reports, bills, feedback and teleconsultation through mobile apps.

**20. Completely Integrated Ecosystem**

HIS, EMR, Pharmacy, Lab, Radiology, Finance, HR, Biomedical, Inventory, Quality and Mobile Apps run on a single platform without third-party dependency.

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## Get in Touch

To request a live demo or discuss a deployment for your hospital, please contact our team.